Java Collections Exercises

1. Create and Display Names

Create an ArrayList of names. Add 5 names and display them using a loop.

2. Insert and Remove Elements

Add elements to a list, remove the 3rd element, and display the remaining items.

3. Sort and Reverse a List

Store integers in a list, sort them in ascending and descending order.

4. Find Duplicates in a List

Given a list of integers, print all duplicate elements.

5. Merge Two Lists

Create two lists and merge them into a single one.

6. Remove Duplicates from a List

Convert a list with repeated values into a set (no duplicates).

7. Compare Two Sets

Create two sets and check if they have any elements in common.

8. Sort a Set

Create a TreeSet of strings and display them in sorted order automatically.

9. Find Union and Intersection

Find the union and intersection of two sets of numbers.

10. Check Element Existence

Create a HashSet of student IDs and check if a particular ID exists.

11. Store Student Marks

Use a HashMap<String, Integer> to store names and marks. Display all entries.

12. Count Word Frequency

Count how many times each word appears in a sentence using a Map.

13. Sort a Map by Keys

Store country–capital pairs and display them in alphabetical order by country.

14. Find the Key with Highest Value

Given a Map<String, Integer>, find the entry with the highest value.

15. 5. Remove Key–Value Pair

Create a HashMap of names and ages, then remove one entry.

16. Sort Objects by Age

Create a Student class (name, age). Store students in a list and sort them by age.

17. Sort by Name Length

Sort a list of strings by their length instead of alphabetically.

18. Sort in Reverse Order

Sort integers in descending order using a custom comparator.

19. Sort Employees by Salary, then Name

Create Employee class (name, salary). Sort first by salary, then by name if equal.

20. Sort Map by Values

Sort a Map<String, Integer> (e.g., student → marks) by its values (marks).